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APPLIED RESEARCH

An AI-Based Decision Support Framework for Clinical Screening of Iron Deficiency Anemia and Thalassemia

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee for Research in Human Subjects at Hatyai Hospital, Thailand, under Approval No. HYH EC 002-67-01, and performed in line with the Declaration of Helsinki.

ABSTRACT Iron deficiency anemia (IDA) and thalassemia (THL) are common hematological disorders that necessitate efficient and accurate screening for early diagnosis. Traditional blood smear analysis is labor-intensive, prone to subjectivity, and lacks reproducibility, highlighting the urgent need for AI-driven methods to improve diagnostic sensitivity and specificity. This study proposes a novel hybrid AI framework for patient-level classification, integrating soft voting with optimized probability-based thresholds. The model was trained and validated using a real-world dataset from Hatyai Hospital, Thailand, and evaluated at both the patch and patient levels. The proposed approach achieved 96% accuracy on the test set. Specifically, it yielded precision-recall values of 1.00 and 0.83 for IDA, and 0.95 and 1.00 for THL, respectively. At the patient level, sensitivity reached 1.00 for THL and 0.83 for IDA. Bayesian probability updates further confirmed prediction reliability, yielding post-test probabilities exceeding 99.99% for IDA and 80% for THL. The model explained 62.84% of the variance in patient classifications, demonstrating strong discriminatory power. Model interpretability, assessed using SHAP and Grad-CAM, highlighted key red blood cell morphological features. The proposed framework thus serves as a cost-effective screening tool. Limitations include the use of a single-center dataset and the need for adaptive threshold optimization. Future work will focus on multi-center validation and real-world clinical integration. This study thereby establishes a structured baseline for AI-assisted hematology screening, fostering early detection and improved clinical decision-making in resource-limited settings.

INDEX TERMS Blood smear, iron deficiency anemia, thalassemia, deep learning, hybrid classification, clinical decision-making, model interpretability.

I. INTRODUCTION

Thalassemia (THL) is a significant health concern on a global scale, affecting approximately 33% of the world's population [1]. According to the World Health Organization (WHO), in 2019, 24% of women aged 15–49 years were diagnosed with anemia, with an even higher prevalence of 32.2% among pregnant women [2]. In Thailand, THL traits are present in 30–40% of the population, with approximately 1% (approximately 660,000 individuals) affected by the disease [3]. Given its high prevalence, early and accurate screening of THL is crucial for effective clinical management. It is imperative that public health professionals prioritize this issue to mitigate the impact of THL on affected individuals and their communities.

Microcytic anemia, characterized by abnormally small red blood cells (RBCs), is primarily attributed to IDA and THL. It is critical to distinguish between these two conditions

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because their treatment approaches differ significantly. The standard diagnostic workflow comprises three key steps: 1) Microscopic examination of RBC morphology by hematologists or laboratory experts; 2) Hemoglobin typing tests to detect abnormal hemoglobin variants; and 3) Iron studies to evaluate blood iron levels [4].

IDA is primarily characterized by iron deficiency, which affects ferritin, serum iron, and transferrin saturation levels. This condition typically presents with high Red Cell Distribution Width (RDW) due to anisocytosis, indicating a variation in RBC size. Treatment with iron supplements is generally effective. In contrast, THL is a genetic disorder that typically exhibits normal or elevated ferritin and iron levels, accompanied by distinctive poikilocytosis, characterized by abnormal RBC shapes such as target cells and tear drop cells. Treatment for THL necessitates hemoglobin typing and iron chelation rather than iron supplementation [1], [5] [6], [7].

Misclassification of IDA as THL, or vice versa, poses significant risks, as inappropriate treatment can have dire consequences. For instance, administering iron supplements to THL patients may lead to iron overload and increase the risk of secondary hemochromatosis, a condition resulting in iron accumulation in vital organs, such as the liver, pancreas, and heart. Conversely, delayed treatment of IDA can lead to severe anemia, cognitive impairment, and diminished quality of life.

Moreover, THL presents severe clinical challenges in pregnant women. If undiagnosed, it can result in Hemoglobin Bart's hydrops fetalis, a fatal condition characterized by severe in utero anemia. This disorder leads to organ enlargement, fluid accumulation, and cardiac failure, often culminating in stillbirth or neonatal death. Given the genetic nature of THL, family-based screening is essential for identifying at-risk individuals and providing appropriate genetic counseling.

Despite the critical need for early diagnosis, rural hospitals and primary healthcare centers frequently lack access to hematologists and laboratory specialists. Consequently, anemia diagnosis often relies on laboratory blood tests, which may necessitate travel to central hospitals, imposing financial burdens and resulting in delayed diagnosis and treatment initiation. Furthermore, the transportation of blood samples and the need for repeated testing exacerbate laboratory costs and prolong waiting times.

A. AI IN IRON DEFICIENCY ANEMIA AND THALASSEMIA DIAGNOSIS

Recent advances in Artificial Intelligence (AI) and Deep Learning (DL) have demonstrated high accuracy and efficiency in medical image classification. Automated blood smear analysis can significantly reduce diagnostic delays and enhance accessibility in resource-limited settings. However, two major challenges persist [8].

1) Data variability and regional differences: AI models trained on Western populations may not generalize well to Southeast Asian cases due to genetic variations in THL mutations [9]. For example, β-thalassemia is common in the Mediterranean, while α-thalassemia is prevalent in Africa and Southeast Asia. This necessitates region-specific adaptation of AI models.
2) Limited clinical deployment: Many AI models for hematology are either proprietary or restricted by licensing agreements, rendering them inaccessible to smaller hospitals. Furthermore, uninterpretable AI models often lack explainability, which can lead to distrust among medical practitioners.

B. STUDY CONTRIBUTION

To address these challenges, the key contributions of this work are as follows:

1) Development of a comprehensive dataset of annotated blood smear images collected from real-world clinical settings in Thailand.
2) Optimization of a DL model for THL and IDA classification, trained using multi-fold cross-validation to ensure robust and reliable performance.
3) Evaluation at both patch and patient levels, ensuring the model's effectiveness in image-based classification and practical clinical decision-making.
4) Implementation of a novel hybrid approach for patient-level classification, designed to maximize sensitivity and minimize the risk of missing THL cases in clinical deployment.
5) Conduct of a comprehensive statistical performance analysis, highlighting both the strengths and limitations of the AI model in real-world clinical settings.

By integrating AI into the early screening workflow, this study aims to reduce diagnostic costs, improve accuracy, and enhance accessibility for patients in under-resourced healthcare facilities. The findings will support clinical decision-making and provide a foundation for deploying AI-based screening tools in hematology.

II. RELATED STUDIES

This section reviews existing works categorized into structured clinical data, medical imaging, and hybrid data integration.

A. AI-BASED CLASSIFICATION USING STRUCTURED CLINICAL DATA

Several studies have explored structured datasets, including blood test results, electronic health records (EHR), and RBC indices for the classification of IDA and THL.

Smith et al. [10] developed a machine learning-based optical scattering system for anemia detection; however, it lacked patient-level probability refinement. Laengsri et al. [11] introduced ThalPred, a web-based classifier for IDA versus THL, but did not incorporate statistical thresholding. Fu et al. [12] and Phirom et al. [13] applied support vector

machines and deep learning for THL classification but failed to integrate hybrid decision-making strategies.

Khorwanichakij et al. [14] proposed a formula-based classification score without integrating soft voting or probability thresholds. Taware et al. [15] and Saputra et al. [16] employed ensemble models demonstrating high accuracy, yet they lacked explainability and patient-level optimization. Lastly, Garduno-Rapp et al. [17] developed an EHR-based IDA prediction models of neural networks that did not emphasize hematology classification.

B. AI FOR HEMATOLOGIC DISORDERS USING MEDICAL IMAGING

Deep learning has been applied to MRI-based THL assessment and blood smear classification.

Zaylaa et al. [18] proposed an AI framework that achieved 88% accuracy for THL diagnosis utilizing MRI segmentation. This framework combined transfer learning (TL) and DL for efficient diagnosis; however, performance evaluation required more extensive dataset testing, and the approach was not applicable to blood smear-based screening.

Lian et al. [19] developed a deep learning model for MRI-based myocardial iron overload assessment, achieving Dice coefficients of 0.90 (internal) and 0.82–0.86 (external) with a dataset of 146 patients from two centers. Their method automates iron overload assessment, thereby reducing operator-dependent variability. Nonetheless, their approach is tailored for MRI-based segmentation of cardiac iron deposition and does not address blood smear-based patient-level hematology classification, which is the focus of our study.

Wiratchawa et al. [20] introduced ThalNet, a convolutional neural network (CNN)-based model trained on blood smear images, achieving an accuracy of 86%, F1-score of 0.85, precision of 0.88, recall of 0.87, and specificity of 0.78. This model utilized a limited dataset of 51 cases and employed EfficientNet-B7 and Vision Transformer (ViT) architectures. However, its performance is highly dependent on high-quality blood smear images and focuses solely on patch-level classification without integrating hybrid probability thresholding for patient-level decision-making.

C. HYBRID APPROACHES INTEGRATING STRUCTURED AND IMAGING DATA

Few studies have successfully integrated clinical data with imaging features for anemia diagnosis. Yadav et al. [21] developed Suśruta, a smartphone-based anemia screening tool that combines complete blood count (CBC) and hemoglobin image data with AI-driven classification. This system utilized structured clinical data (CBC and liquid chromatography) and image analysis (hemoglobin images) from 60,283 cases collected over eight years, employing machine learning (ML), DL, Bayesian Knowledge Networks, and Computer Vision for predictive modeling. However, the study's approach lacked probability-based thresholding for decision-making, which limited its interpretability in patient-level classification.

D. SUMMARY OF CONTRIBUTIONS

Prior studies have primarily focused on structured ML models, medical imaging AI, or hybrid AI applications; however, they generally lack hybrid probability thresholds, statistical validation, and explainability for patient-level hematology screening.

This study integrates soft voting with optimized probability thresholds, including mean-based, argmax-based, and percentile-based cutoffs, to enhance decision robustness and statistical validation. Furthermore, our method incorporates explainability through SHAP (SHapley Additive exPlanations), Grad-CAM (Gradient-weighted Class Activation Mapping), and post-test probability analysis.

III. MATERIALS AND METHODS

This section describes the study design, dataset, proposed methodology, and deep learning architecture used in this study.

A. DATASET

All procedures conducted in this study complied with the Declaration of Helsinki and received approval from the Ethics Committee for Research in Human Subjects at Hatyai Hospital, Thailand (HYH EC 002-67-01). This approval ensured adherence to International Conference on Harmonisation-Good Clinical Practice (ICH-GCP) guidelines and addressed all pertinent ethical considerations. The characteristics of the study population, along with the respective inclusion and exclusion criteria, are reported in Table 1.

1) DATA ACQUISITION

Peripheral blood smear (PBS) slides were selected based on the red blood cell morphologies relevant to IDA and THL classification, as illustrated in Table 2.

The collected slides were prepared and analyzed using the Sysmex XN-300^{TM} Automated Hematology Analyzer (Hamburg, Germany). The Leica ICC50W digital microscope, equipped with binocular vision, was utilized to inspect the blood smear slides and capture high-resolution digital images.

To ensure slide quality, staining consistency and the presence of non-overlapping RBCs were assessed prior to microscopic examination. RBC morphology was then inspected at 100X magnification. For each PBS slide, ten images were captured at different locations, maintaining the same magnification. The final images were saved in JPG format with a resolution of 1,920 × 1,080 pixels in the RGB color space.

2) DATA LABELING

The labeling process, also referred to as ground truth annotation, was conducted by integrating the collected PBS

TABLE 1. Study population and inclusion/exclusion criteria.

Attribute	Detail
Target Population	Patients diagnosed with anemia
Study Population	Patients diagnosed with THL, THL trait, or IDA at Hatyai Hospital
Inclusion Criteria	Age $\geq$ 18 years old
	Confirmed diagnosis of THL, THL trait, or IDA
	Persistent anemia at the time of participation
Exclusion Criteria	Anemia caused by other conditions: aplastic anemia, anemia of chronic disease, autoimmune hemolytic anemia, or multiple myeloma or leukemia.
	Co-occurrence of both THL and IDA
	Chronic kidney disease (CKD) stage 3 or higher (eGFR < 60 ml/min/1.73 m^2)
Sample Size	150 blood smear slides per study group (THL, THL trait, IDA)
Total Slides	450 slides

TABLE 2. Red blood cell morphologies relevant to IDA and THL classification.

RBC Morphology	Size	Hypochromic Microcytic
Normal RBC	Normocytic	No
Microcytic RBC Variants*	Microcytic	Yes (IDA, THL)
Macrocytes	Macrocytic	No (Seen in B12/Folate Deficiency)
Target Cell*	Microcytic	Yes (THL, Liver Disease)
Tear Drop Cell*	Microcytic	Yes (THL, Myelofibrosis)

* Indicates red blood cell morphologies commonly associated with microcytic anemia, including IDA and THL.

TABLE 3. Anemia definition, sourced WHO.

Gender	Hemoglobin (gm/dL)	Hematocrit (%)
Male	< 13	< 39
Female	< 12	< 36
Pregnant Women	< 11	< 33

TABLE 4. Diagnostic criteria for IDA and THL.

Criteria	IDA	THL
Red Blood Cell Morphology	Hypochromic microcytic	Hypochromic microcytic with poikilocytosis
Serum Iron	< 7.1 mg/L	Normal or High
Serum Ferritin	< 30 ng/L	Normal or High
Transferrin Saturation	< 15%	Normal
Total Iron-Binding Capacity (TIBC)	> 13.1 μmol/L	Normal
Hemoglobin Typing	Normal	Abnormal (consistent with trait or disease)

slide images with CBC test results and hematologists' diagnoses.

CBC tests were employed to identify microcytic anemia, characterized by small red blood cells, utilizing a mean corpuscular volume (MCV) threshold of < 80 fL. The study adhered to the WHO anemia criteria [22], [23], as summarized in Table 3.

Hematologists utilized CBC test results in conjunction with additional diagnostic assessments to classify cases into three categories: IDA, THL trait, and THL disease. The key diagnostic criteria for distinguishing between IDA and THL were derived from Johnson-Wimbley et al. [4] and are outlined in Table4. For the development of the AI model in this study, we focused exclusively on binary classification—IDA and THL—as our primary priority for clinical screening.

3) DATA PREPROCESSING

Following data acquisition, we processed and summarized the study population into two categories, as presented in Table 5. However, due to image damage, the final study population was reduced to 111 cases, comprising 44 IDA cases and 67 THL cases.

The preprocessing steps applied to the blood smear images are illustrated in Figure 1. The images were stain-normalized to the LAB color space using the method proposed by Shen et al. [24].

TABLE 5. Demographic table for THL and IDA.

Variable	Label	IDA (%)	THL (%)
Study population	Anemia patients	N = 46 (35.1)	N = 85 (64.9)
Sex	Male	11 (23.9)	36 (42.4)
	Female	35 (76.1)	49 (57.6)
Age	Min	16	18
	Max	94	89
	Mean ± SD	46.24 ± 21.13	44.36 ± 17.04
Hb (gm/dL)	Mean ± SD	6.84 ± 1.85	7.63 ± 1.70
Hct (%)	Mean ± SD	22.84 ± 5.28	24.58 ± 4.68
WBC (cells/μL)	Mean ± SD	9,455 ± 3,958	9,033 ± 4,723
Platelet (cells/μL)	Mean ± SD	359,587 ± 130,323	303,889 ± 171,074
MCV (fL)	Mean ± SD	64.53 ± 9.04	64.15 ± 6.59
MCH (pg)	Mean ± SD	19.52 ± 4.86	20.03 ± 2.47
MCHC (g/dL)	Mean ± SD	29.38 ± 3.00	31.01 ± 2.76

Hb = Hemoglobin, Hct = Hematocrit, MCV = Mean Corpuscular Volume, MCH = Mean Corpuscular Hemoglobin, MCHC = Mean Corpuscular Hemoglobin Concentration.

Each image (region-level per patient) was divided into cell-level patches of size 224 × 224 pixels, as detailed in Algorithm 1. A pixel ratio threshold of greater than 50.0% was applied to each patch to ensure quality control (QC). All patches were manually inspected to remove artifacts and damaged cells, with only QC-approved patches included in the training and testing sets.

For the unseen testing set, equal samples were allocated from six patients with IDA and six with THL (a total of 12 subjects), representing approximately 11% of the total 111 cases. The remaining 99 cases formed the training set, which was further divided into five-fold cross-validation sets using a 75:25 ratio, as summarized in Table 6.

To enhance dataset diversity, image augmentation techniques, including horizontal and vertical flipping, were applied. Additionally, to address class imbalance, we randomly oversampled patches from the minority class until it matched the majority class, ensuring a balanced dataset [25].

Figure 2 illustrates the number of samples in each fold after augmentation and balancing. The total number of patches ranged from 126,018 to 159,918 for the training set and from 34,818 to 54,816 for the validation set. For the balanced test set, the total number of patches was 14,898, with each class containing 7,449 patches.

Example image patches from the balanced dataset, representing both IDA and THL cases across all splits, are shown in Figure 3 in RGB color mode. During training, patches were converted from RGB to BGR to match the input format required by OpenCV-based pipelines. This step followed stain normalization performed in the LAB color space to ensure consistency in color representation across samples.

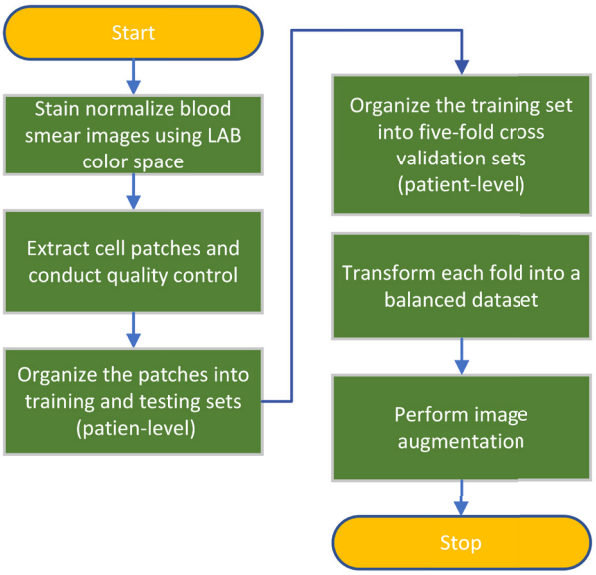


FIGURE 1. Preprocessing steps for blood smear images, illustrating the generation of balanced datasets for five-fold cross-validation.

B. STUDY DESIGN AND EVALUATION METRICS

The term *TP* represents True Positives, which are the number of correctly predicted positive cases (THL or 1). The term *TN* stands for True Negatives, referring to the number of correctly predicted negative cases (IDA or 0). The term *FP* denotes False Positives, which occur when actual negative cases are incorrectly classified as positive. Similarly, *FN* stands for False Negatives, which occur when actual positive cases are misclassified as negative [26].

Algorithm 1 Algorithm for Extracting Image Patches at the Cell Level

Require: *image_files* ▷ List of input blood smear images in region-level
Require: $min_area = 512$, $target_size = 224$, $ratio = 50.0$
Ensure: *isolated_cells_qc* ▷ List of passed quality control image patches in cell-level

for *image_path* ∈ *image_files* **do**
  image ← load_image(*image_path*)
  gray ← convert_to_grayscale(*image*)
  blurred ← apply_gaussian_blur(*gray*, (5, 5))
  thresholded ← apply_threshold(*blurred*, 140)
  contours ← find_contours(*thresholded*)
  Initialize empty lists *isolated_cells*, *isolated_cells_qc*
  for *contour* ∈ *contours* **do**
    if contourArea(*contour*) > *min_area* **then**
      (x, y, w, h) ← boundingRect(*contour*)
      if $x > 5$ and $y > 5$ and $(x + w) < image_width - 5$ and $(y + h) < image_height - 5$ **then**
        $padding_top \leftarrow 30$, $padding_bottom \leftarrow 25$
        $w \leftarrow w + 30$, $h = h + padding_top + padding_bottom$
        $x \leftarrow x - 15$, $y \leftarrow y - padding_top$
        Ensure x, y, w, h are within image bounds
        Append (x, y, w, h) to *isolated_cells*
      end if
    end if
  end for
  for *index*, *cell* ∈ *enumerate*(*isolated_cells*) **do**
    $cropped_cell_{index}$ ← crop_cell(*index*, *image*, *cell*)
    if $cropped_cell_{index} \neq None$ **then**
      pix_ratio ← convertBinary($cropped_cell_{index}$)
      if $pix_ratio \geq ratio$ **then**
        Append $cropped_cell_{index}$ to *isolated_cells_qc*
        Convert $cropped_cell_{index}$ to RGB
        Save_image $cropped_cell_{index}$
      end if
    end if
  end for
end for

TABLE 6. Five-fold cross-validation dataset splits, detailing the composition of training and validation sets.

Fold	No. of Subjects						No. of Patches					
	Training Set			Validation Set			Training Set			Validation Set		
	Total	**IDA**	**THL**	**Total**	**IDA**	**THL**	**Total**	**IDA**	**THL**	**Total**	**IDA**	**THL**
0	79	25	54	20	13	7	40,303	14,257	26,046	11,003	6,910	4,093
1	79	33	46	20	5	15	40,899	18,630	22,269	10,407	2,537	7,870
2	79	28	51	20	2	18	41,504	19,735	21,769	9,802	1,432	8,370
3	79	28	51	20	10	10	39,731	14,844	24,887	11,575	6,323	5,252
4	80	30	50	19	8	11	42,787	17,202	25,585	8,519	3,965	4,554

In this study, we defined a standard baseline for correctly identifying cases with a probability $\geqslant 80\%$. We defined the set of evaluation metrics, $\mathcal{M}$, as presented in Equations 1 to 6 [27], [28] for the unseen testing set, $\mathcal{D}_{test}$.

Precision focuses on the proportion of predicted positives that are correct; thus, False Positives (FP) are particularly costly, as they may lead to unnecessary treatments or further diagnostics. The formula for precision (also known as Positive Predictive Value, PPV) is given by Equation 2, where C is the total number of classes in the classification task. This formula calculates the mean sensitivity across all C classes.

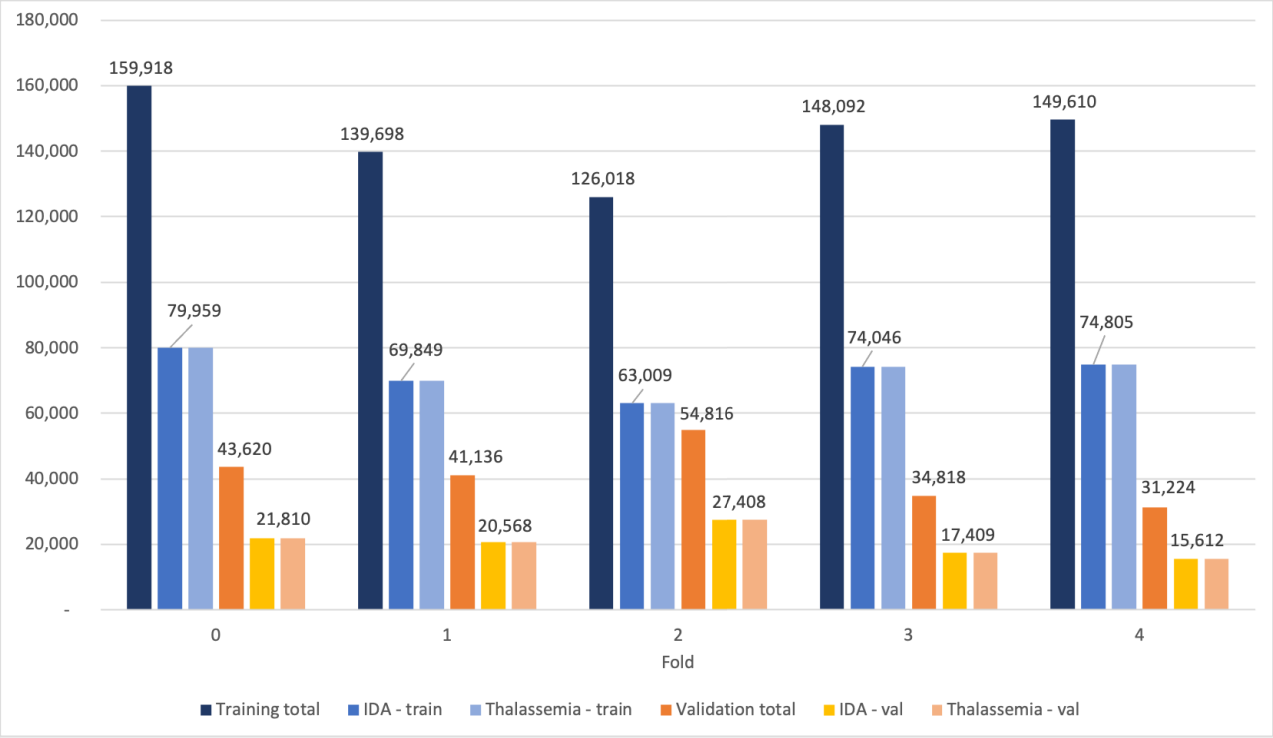


FIGURE 2. Five-fold cross-validation of the dataset after augmentation and balancing, achieved by randomly oversampling minority class patches until they matched the majority class.

Analogously, Negative Predictive Value (NPV) measures the proportion of predicted negatives that are actually correct. NPV is crucial when False Negatives (FN) are costly, such as in early-stage cancer screening, where missing a diagnosis can have severe consequences. The formula for NPV is provided by Equation 3.

Sensitivity measures the proportion of actual positive cases (THL) that are correctly identified by the model, as False Negatives (FN) can be dangerous for patients, leading to missed necessary treatments. The formula for sensitivity (Recall or True Positive Rate, TPR) is given by Equation 4.

For IDA classification (Class 0), specificity (True Negative Rate, TNR), as defined in Equation 5, is critical in preventing THL misclassification and avoiding incorrect iron therapy. High specificity ensures that THL cases are not misdiagnosed as IDA, supporting safe and effective anemia screening, ultimately leading to improved clinical outcomes.

Additionally, the F1-score, which represents the harmonic mean of precision and sensitivity, is defined in Equation 6. Given that accuracy alone can be misleading in an imbalanced dataset, the F1-score was selected to mitigate the impact of class imbalance, providing a more reliable performance evaluation.

1) PATCH-LEVEL PERFORMANCE METRICS

This equation is particularly useful in multi-class classification, where the sensitivity of each class can be calculated, such as Recall_0 for correctly detected IDA and Recall_1 for correctly detected THL. In addition, the average sensitivity across all classes is used to evaluate the model's overall ability to correctly identify cases.

Let $F = 5$ be the number of folds in a cross-validation dataset, and let $\mathcal{D}_{\text{test}}$ denote the test dataset for each fold. The set $\mathcal{M}$ includes the evaluation metrics defined in Equations 1–6.

For a single model, the average value of each evaluation metric m across all folds is computed using Equation 7, where $S_{f,i}(m, \mathcal{D}_{\text{test}})$ represents the metric value m evaluated for model i on fold f.

Since we already have $S_{avg}(m)$ defined for each metric m across five folds, now we need to compute the mean across all metrics for each model as defined in Equation 8. Finally, we select the best model performance based on the highest $S_{final,i}$, as defined in Equation 9.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$PPV = \frac{1}{C}\sum_{j=0}^{C-1}\frac{TP_j}{TP_j + FP_j}, \quad j \in \{0, 1\} \tag{2}$$

$$NPV = \frac{1}{C}\sum_{j=0}^{C-1}\frac{TN_j}{TN_j + FN_j}, \quad j \in \{0, 1\} \tag{3}$$

$$Sensitivity = \frac{1}{C}\sum_{j=0}^{C-1}\frac{TP_j}{TP_j + FN_j}, \quad j \in \{0, 1\} \tag{4}$$

$$Specificity = \frac{1}{C}\sum_{j=0}^{C-1}\frac{TN_j}{TN_j + FP_j}, \quad j \in \{0, 1\} \tag{5}$$

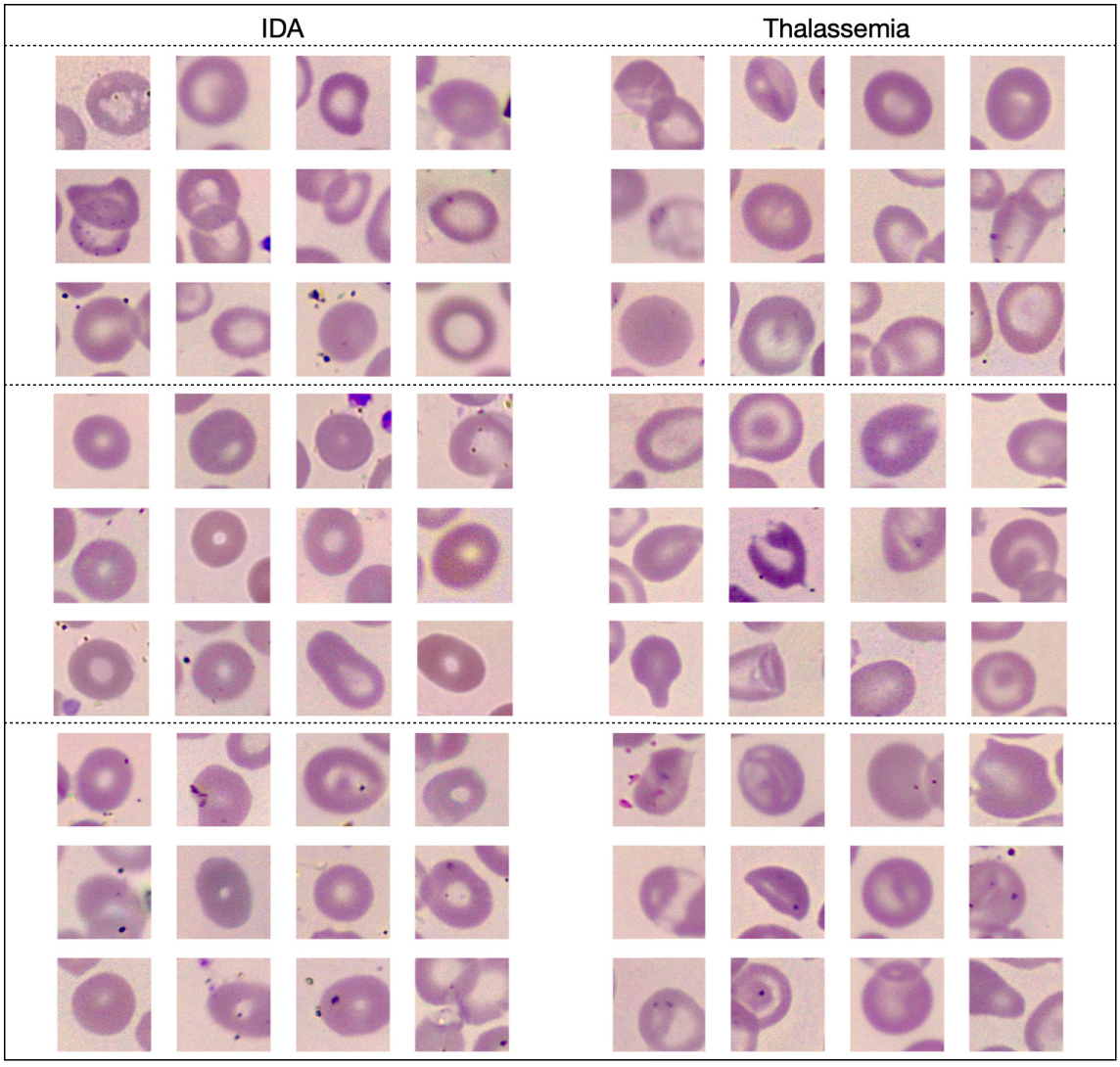


FIGURE 3. Example patches from the balanced dataset of IDA and THL. The top row represents the training set, the middle row corresponds to the validation set, and the bottom row illustrates the testing set.

$$F1_{score} = \frac{1}{C}\sum_{j=0}^{C-1}\frac{2 \times \mathrm{PPV}_j \times \mathrm{TPR}_j}{\mathrm{PPV}_j + \mathrm{TPR}_j}, \quad j \in \{0, 1\} \tag{6}$$

$$S_{avg,i}(m) = \frac{1}{F}\sum_{f=0}^{F-1} S_{f,i}(m, \mathcal{D}_{test}), \quad m \in \mathcal{M} \tag{7}$$

$$S_{final,i} = \frac{1}{|\mathcal{M}|}\sum_{m\in\mathcal{M}} S_{avg,i}(m) \tag{8}$$

$$S_{\max} = \max_{i\in\{0,1,2,3,4\}} S_{\text{final},i} \tag{9}$$

$$\bar{p}_{i,j}(\mathcal{D}^f_{val}) = \frac{1}{M}\sum_{m=1}^{M} p^m_{i,j}(\mathcal{D}^f_{val}), \quad j \in \{0, 1\}, \quad m \in \mathcal{M}_f \tag{10}$$

$$P^f_j = \frac{1}{N^f}\sum_{i=1}^{N^f}\bar{p}_{i,j}, \quad j \in \{0, 1\} \tag{11}$$

$$\hat{P}^f_{1,\text{mean-based}} = \frac{1}{N^f}\sum_{i=1}^{N^f}\bar{p}_{i,1} \tag{12}$$

$$\hat{y}^f_i = \arg\max_j(p_{i,j}), \quad j \in \{0, 1\} \tag{13}$$

$$\hat{P}^f_{1,\text{argmax-based}} = \frac{\sum_{i=1}^{N^f} 1(\hat{y}^f_i = 1)}{N^f} \tag{14}$$

$$T^*_{\text{patient,method},f} = \arg\max_T \left(\text{Sensitivity}^f(T) + \text{Specificity}^f(T) - 1\right),$$

$$T^*_{\text{patient,method}} = \frac{1}{F}\sum_{f=0}^{F-1} T^*_{\text{patient,method},f},$$

$$\text{method} \in \{\text{mean-based, argmax-based}\} \tag{15}$$

$$\hat{Y}_{\text{patient}}(\mathcal{D}_{test}) = \begin{cases} 1, & \text{if } \hat{P}_{1,\text{method}}(\mathcal{D}_{test}) \geq T^*_{\text{patient,method}} \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

2) CROSS-VALIDATION INFERENCE WITH UNSEEN VALIDATION FOLDS FOR PATIENT-LEVEL CLASSIFICATION

For patient-level performance evaluation, we employed two methods: mean-based probability and argmax-based probability, utilizing an ensemble of five trained models. The final classification decision at the patient level was derived by aggregating patch-level predictions across these models. During the evaluation phase, optimal thresholds for both methods were determined using the validation folds and subsequently applied during inference, as outlined in Algorithm 2. The process follows these steps:

1) Patch-Level Probability Aggregation
   Equation 10 computes the average predicted probability $p_{i,j}$ for class j at the patch level. This is achieved by aggregating predictions from M_f models trained on different folds and computing their mean. This step enhances robustness and reduces bias across models.
2) Patient-Level Probability Estimation
   Equation 11 calculates the patient-level probability P_j^f, where N^f represents the total number of patches for a patient in validation fold f. It averages all patch-level probabilities $\bar{p}_{i,j}$, ensuring a proportional contribution from all patches to the final classification decision.
3) Mean-Based Classification Probability for THL
   Equation 12 is a special case of Equation 11 for Class 1 (THL). It estimates the likelihood of a patient being classified as THL by averaging the soft probabilities $\bar{p}_{i,1}$ from all patches of that patient.
4) Patch-Level Hard Classification via Argmax
   Equation 13 assigns a discrete class label to each patch i by selecting the class j with the highest predicted probability. This follows the standard argmax decision rule used in classification tasks.
5) Patient-Level Hard Voting Classification
   Equation 14 applies hard voting at the patient level. It computes the proportion of patches classified as THL (Class 1) among all patches for a given patient. The indicator function $1(\hat{y}_i^f = 1)$ counts the patches classified as THL, and the fraction normalizes it across all patient patches.

During the evaluation phase, the optimal threshold was selected by maximizing Youden's Index, defined as Sensitivity + Specificity - 1. Originally introduced by Youden in 1950 [29], this metric is widely used for determining the optimal cutoff in classification models, particularly in Receiver Operating Characteristic (ROC) curve analysis [30].

We developed an algorithm to compute the optimal threshold $T^*_{\text{patient,method},f}$ for patient classification, which differs from the patch-level threshold for each method. For each validation fold, the optimal threshold was derived by maximizing Youden's Index, as shown in Equation 15. The final threshold, $T^*_{\text{patient,method}}$, was obtained by averaging across all validation folds.

During inference, the patient probability $P_{\text{patient}}(\mathcal{D}_{test})$ was compared against the optimal threshold $T^*_{\text{patient,method}}$ as defined in Equation 16. If $P_{\text{patient}}(\mathcal{D}_{test}) \geq T^*_{\text{patient,method}}$, the patient was classified as THL (Class 1); otherwise, the patient was classified as IDA (Class 0). Selecting an appropriate method for aggregating patch-level predictions into patient-level decisions is crucial for clinical applicability. The experimental results and analysis are further discussed in Section IV.

C. CLASSIFICATION METHOD USING TRANSFER LEARNING TECHNIQUE

In this work, we employed the TL technique, recognized for its simplicity and efficiency [31] [32] [33]. The candidate pretrained models were categorized into two primary groups: ViT-based models [34] and CNN-based models [35].

For the ViT-based models, we utilized the Gaussian Error Linear Unit (GeLU) activation function [36], as defined in Equations 17 and 18. It multiplies the input x by the cumulative distribution function (CDF) of the standard normal distribution, denoted as $\Phi(x)$. Conversely, for the CNN-based models, the Rectified Linear Unit (ReLU) activation function [37], [38], as defined in Equation 19, was employed. This approach facilitates effective feature extraction from the data, leveraging the strengths of both model types for enhanced classification performance.

The candidate models were constructed by integrating its respective backbone with custom layers, which included fully connected (FC) layers consisting of 1024, 512, 256, 64, 32, and 16 neurons. To prevent overfitting, dropout (rate = 0.4) [38] and batch normalization were applied to each fully connected layer. Notably, L2 decay was implemented solely on the first layer to regularize the model. The final prediction layer employed the Softmax activation function [39], as defined in Equation 20, to facilitate binary classification.

The candidate models were compiled using the Adam optimizer [40], which incorporates both the learning rate and weight decay, both set to 1×10^{-4}. The binary cross-entropy loss function [41], as defined in Equation 21, was employed. In this context, y represents the true label, while $\hat{y}$ denotes the predicted probability (ranging between 0 and 1) for a single sample. This configuration enables the model to effectively learn from the data by minimizing the prediction error during training.

$$\text{GeLU}(x) = x \cdot \Phi(x) \tag{17}$$

$$\Phi(x) = \frac{1}{2}\left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}\left(x + 0.044715\, x^3\right)\right)\right) \tag{18}$$

$$g(\vec{z}) = max(0, z) \tag{19}$$

$$\sigma(\vec{z})_j = \frac{e^{z_j}}{\sum_{k=1}^{K} e^{z_k}} \tag{20}$$

$$\mathcal{L}_{\text{BCE}}(y, \hat{y}) = -\left[y \log \hat{y} + (1 - y) \log(1 - \hat{y})\right] \tag{21}$$

The learning rate was determined from experimental results using Algorithm 3, resulting in an optimal value

Algorithm 2 Cross-Validation Inference With Unseen Validation Folds for Patient-Level Classification

Require: Trained ensemble models; $models_{\text{trained}}$, five-fold cross-validation sets; $\mathcal{D}_{val}$
Ensure: Patient-level predictions; P_{patient}

for $(val_fold, val_set) \in \mathcal{D}_{val}$ **do** ▷ Iterate through each validation fold
  $train_folds \leftarrow \{0, 1, 2, 3, 4\} \setminus \{val_fold\}$ ▷ Select training folds
  $models_{\text{used}} \leftarrow \{models[i] \mid i \in train_folds\}$ ▷ Select trained models excluding validation fold
  for $(patient_id, class_id) \in val_set$ **do**
    $patch_files \leftarrow$ find_patches$(patient_id, class_id)$
    $patient_df \leftarrow$ create_dataframe$(patch_files, class_id)$
    Initialize $y_test_pred_ensemble$ and $y_test_pred_each_argmax_list$
    for $model \in models_{\text{used}}$ **do** ▷ Predict using all trained models
      for $batch \in patient_gen$ **do**
        $y_test_pred \leftarrow$ predict$(model, batch)$
        Store y_test_pred in $y_test_pred_ensemble$
        Store argmax(y_test_pred) in $y_test_pred_each_argmax_list$
      end for
    end for
    $patch_prob_avg \leftarrow$ mean$(y_test_pred_ensemble, axis = 0)$
    $P_{\text{patient,mean}} \leftarrow$ mean$(patch_prob_avg[:, 1])$
    $y_test_pred_argmax \leftarrow$ argmax$(patch_prob_avg, axis = 1)$
    $(u, c) \leftarrow$ unique$(y_test_pred_argmax, return_counts = True)$
    $class_1_count \leftarrow c[u == 1][0]$**if**$1 \in u$**else**$0$
    $total_patches \leftarrow$ sum(c)
    $P_{\text{patient,argmax}} \leftarrow class_1_count / total_patches$
    Store $\{probability_1 : P_{\text{patient,mean}}, class : class_id\}$ in $patient_predictions_{\text{mean}}$
    Store $\{probability_1 : P_{\text{patient,argmax}}, class : class_id\}$ in $patient_predictions_{\text{argmax}}$
  end for
  Append $patient_predictions_{\text{mean}}$ to $all_patient_predictions_{\text{mean}}$
  Append $patient_predictions_{\text{argmax}}$ to $all_patient_predictions_{\text{argmax}}$
end for

of 1×10^{-4}. We conducted exhaustive experiments on the candidate models, initializing ImageNet weights [42] solely for the respective base model. The training process was configured with a batch size of 32 and a total of 120 training epochs, incorporating an early stopping patience of 12 epochs. Given the small learning rate, we applied a learning rate reduction patience of 2, with a reduction factor of 0.5. This configuration aimed to enhance model convergence while mitigating the risk of overfitting during the training process.

IV. RESULTS AND DISCUSSION

This section presents the results of our proposed hybrid classification framework, evaluated at both patch and patient levels. We analyze patch-level model performance across various architectures, determine optimal decision thresholds, and introduce a hybrid approach to enhance patient-level classification robustness. Performance metrics are evaluated, followed by statistical validation to support clinical applicability. Model interpretability is assessed using SHAP and Grad-CAM visualizations. Finally, we compare our findings to prior studies to highlight contributions and identify potential improvements.

A. PATCH-LEVEL CLASSIFICATION PERFORMANCE

We evaluated five candidate models: ViT_b32, EfficientNetV2S, Xception, ResNet50, and ResNet50V2, assessing their patch-level performance using Equation 9. Among these, EfficientNetV2S achieved the highest performance, with an average F1-score of 0.80 for THL and 0.77 for IDA, surpassing ViT-b32 (F1-score = 0.76) and ResNet50V2 (F1-score = 0.75).

Given its superior performance, EfficientNetV2S was selected for further analysis. To ensure a robust evaluation, we applied Equation 22 across all five trained models of EfficientNetV2S. This assessment included key performance metrics—TPR, TNR, PPV, NPV, and F1-score—calculated separately for IDA and THL. We computed the minimum, maximum, and average values across five cross-validation folds, where M_0 to M_4 represent the respective metric values from Fold-0 to Fold-4.

These computations provide a comprehensive assessment of the model's consistency and generalization across folds, as illustrated in Figure 4.

$$f(M, \mathcal{D}_{test}) = \left(\min_{i \in \{0,1,2,3,4\}} M_i, \max_{i \in \{0,1,2,3,4\}} M_i, \frac{1}{5} \sum_{i=0}^{4} M_i \right) \tag{22}$$

Algorithm 3 Algorithm for Range Test Tuning to Determine the Optimal Learning Rate for Model Training

Require: *initial_lr* ▷ Initial learning rate (e.g., $1e^{-5}$)
Require: *desired_lr* ▷ Target learning rate (e.g., 1)
Require: *num_steps* ▷ Total number of iterations for the test (e.g., 120)
Require: *plotfile* ▷ Filename for the plot

Define class `LRTensorBoard` **inherits** `Callback`
  Initialize:
    $lr_multiplier \leftarrow \left(\frac{desired_lr}{initial_lr}\right)^{\frac{1}{num_steps}}$
    $iterations \leftarrow 0$, $lrs \leftarrow []$, $losses \leftarrow []$
  Function `on_train_batch_end`(*batch*, *logs*)
    $lr \leftarrow initial_lr \times (lr_multiplier)^{iterations}$
    lrs.append(*lr*)
    losses.append(*logs_loss*)
    $iterations \leftarrow iterations + 1$
  Function `on_epoch_end`(*epoch*, *logs*)
    If $iterations = 0$ **then**
      return ▷ No training data collected
    End If
    Create a plot with *lrs* on a logarithmic x-scale and *losses* on the y-scale
    Save the plot as *plotfile*

batch_size ← 8
initial_lr ← $1e^{-5}$
desired_lr ← 1
num_steps ← 120
plotfile ← "lr_range_test_plot_1.png"
lr_range_test ← LRTensorBoard(*initial_lr*, *desired_lr*, *num_steps*, *plotfile*)
optimizer ← Adam(*learning_rate* = *initial_lr*)
model.compile(*optimizer*, *loss* = "binary_crossentropy", *metrics* = ["accuracy"])
model.fit(*training_data*, *validation_data*, *epochs* = 1, *callback* = *lr_range_test*)

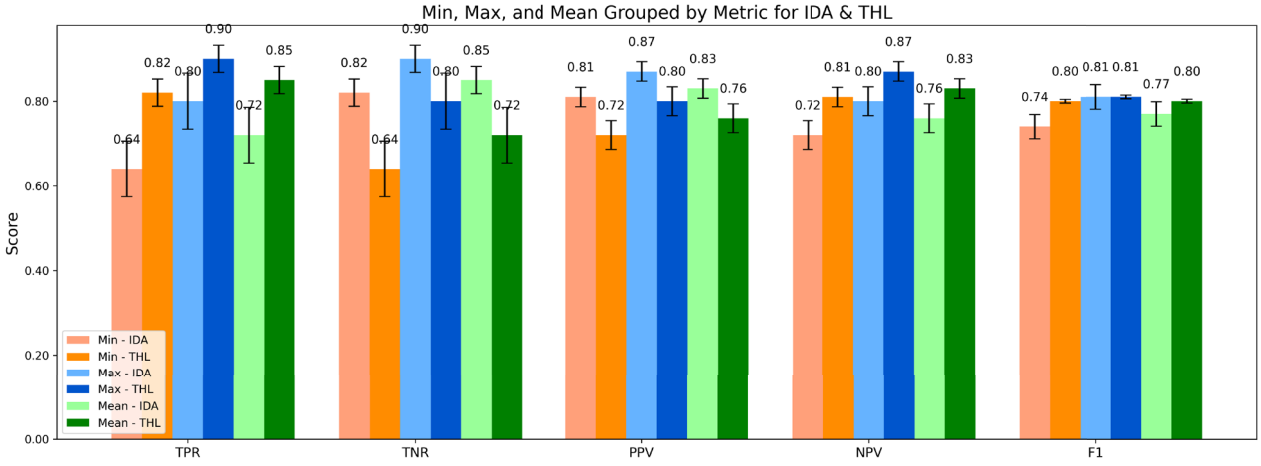


FIGURE 4. Performance evaluation for the testing set in patch-level using the best performance ensemble models of EfficientNetV2S, grouped by metric. The models obtained average F1-score of 0.77 and 0.80 for IDA and THL respectively.

Focusing on the mean values, the models based on EfficientNetV2S demonstrated superior performance in detecting THL cases, achieving a higher TPR of 0.85 compared to a TPR of 0.72 for IDA cases. However, the models encountered challenges with FP for THL, while exhibiting greater confidence in ruling out IDA cases, reflected in a TNR of 0.85. The PPV of 0.83 indicated that IDA predictions were more reliable than those for THL, which had a PPV of 0.77.

Overall performance, as indicated by the F1-score, suggested that the models were slightly better at classifying THL cases (F1-score = 0.80) than IDA cases (F1-score = 0.77).

B. ESTABLISHMENT OF THE OPTIMAL THRESHOLD

To detect every possible THL case, high sensitivity (recall) is prioritized to ensure nearly all actual cases are identified while minimizing excessive confirmatory tests (false positives).

Youden's Index, a metric widely used for determining the optimal cutoff in classification models [29], is employed to determine the optimal threshold $T^*_{patient}$ to achieve the best trade-off between sensitivity and specificity (Equation 15). Higher thresholds favor precision by reducing false positives but may increase false negatives, whereas lower thresholds enhance recall but may introduce more false positives.

Since predictions were made at the patch level, patient-level classification was determined using mean-based and argmax-based probability thresholds, which aggregate patch-level predictions for each patient (Algorithm 2 and Equation 16).

Despite applying data augmentation, the dataset remained THL-dominant. To mitigate overfitting, Youden's Index was averaged across folds for a more reliable and unbiased threshold selection. A mean-based threshold of 0.593 was applied to ensure prediction stability, classifying any probability $\geqslant$ 0.593 as THL (Class 1) to optimize sensitivity and specificity. An argmax-based threshold of 0.634 was also utilized to enhance prediction confidence, closely aligning with the mean-based threshold.

While both thresholds provided stable classification, a hybrid approach was introduced to refine cases within the uncertain probability range, ensuring a more confident final decision.

C. HYBRID APPROACH FOR PATIENT-LEVEL CLASSIFICATION

For clinical deployment, achieving both high sensitivity and interpretability is essential. The hybrid approach refines patient-level classification by addressing discrepancies between mean-based and argmax-based predictions, thereby ensuring stable and reliable outcomes.

To enhance adaptability, we introduced a percentile-based hybrid strategy that integrates percentile-based thresholds for classification refinement. This approach improves robustness while maintaining predictive stability.

The hybrid method defines an uncertain probability range within the mean-based approach. If a patient-level probability falls within this range, the classification is refined using the argmax-based method, providing a more confident and interpretable decision.

Let:

- P_{patient} denote the patient-level probability score, derived from the mean-based method.
- T_{upper} denote the upper bound, the optimal patient-level threshold obtained from the argmax-based method.
- T_{lower} denote the lower bound, computed using a percentile-based threshold.
- $\hat{Y}_{\text{patient}}$ denote the final classification decision.

$$\hat{Y}_{\text{patient}} = \begin{cases} 1, & \text{if } P_{\text{patient}} \geq T_{\text{upper}} \\ \arg\max\limits_{j\in\{0,1\}} \sum\limits_{i=1}^{N} \mathbf{1}(\hat{y}_i = j), & \text{if } T_{\text{lower}} \leq P_{\text{patient}} < T_{\text{upper}} \\ 0, & \text{if } P_{\text{patient}} < T_{\text{lower}} \end{cases} \tag{23}$$

where $\mathbf{1}(\hat{y}_i = j)$ is an indicator function:

$$\mathbf{1}(\hat{y}_i = j) = \begin{cases} 1, & \text{if } \hat{y}_i = j \\ 0, & \text{otherwise} \end{cases} \tag{24}$$

To ensure that each validation fold remains unseen while the other four folds are utilized for training, we define the training-validation split as follows:

$$f_{\text{train}} = \{0, 1, 2, 3, 4\} \setminus f_{\text{val}}, \quad f_{\text{val}} \in \{0, 1, 2, 3, 4\} \tag{25}$$

Each fold serves as a validation set in turn, allowing for comprehensive model evaluation across all available data. Specifically, one fold is held out as the validation set, while the remaining four folds are combined to form the training set. This method allows for a robust assessment of the model's performance, facilitating the identification of any potential overfitting and ensuring that the model generalizes well to unseen data.

The upper bound threshold T_{upper} is derived using Youden's Index to maximize both sensitivity and specificity across the five-fold validation data.

$$T_{\text{upper},f_{\text{val}}} = \arg\max_{T} \left(\text{Sensitivity}_{f_{\text{val}}}(T, f_{\text{train}}) + \text{Specificity}_{f_{\text{val}}}(T, f_{\text{train}}) - 1\right) \tag{26}$$

By averaging across all folds, the final threshold is obtained as:

$$T_{\text{upper}} = \frac{1}{F-1} \sum_{f_{\text{val}}=0}^{F-1} T_{\text{upper},f_{\text{val}}} \tag{27}$$

Similarly, the lower bound threshold T_{lower} is determined using a percentile-based approach:

$$T_{\text{lower},f_{\text{val}}} = P_{\text{patient},(10\%\text{ percentile, fold } f_{\text{val}})} \tag{28}$$

$$T_{\text{lower}} = \frac{1}{F-1} \sum_{f_{\text{val}}=0}^{F-1} T_{\text{lower},f_{\text{val}}} \tag{29}$$

The hybrid approach classifies patients based on the following decision rules:

- High-confidence positive classification ($P_{\text{patient}} \geq 0.593$): The patient is classified as THL (Class 1) with high certainty, maximizing sensitivity and specificity while reducing false negatives.

- Uncertain probability region ($0.260 \leq P_{\text{patient}} < 0.593$): The final decision is determined via majority voting, where the predicted class corresponds to the most frequently occurring patch-level classification. This step refines borderline cases, ensuring greater interpretability and robustness.
- High-confidence negative classification ($P_{\text{patient}} < 0.260$): The patient is classified as IDA (Class 0) with high confidence, minimizing the risk of false positives.

The final hybrid classification framework is formally defined as:

$$\hat{Y}_{\text{patient}}(\mathcal{D}_{test}) = \begin{cases} 1, & \text{if } P_{\text{patient}} \geq 0.593 \\ \arg\max\left(\sum_{i=1}^{N} \mathbf{1}(\hat{y}_i = j)\right), & \text{if } 0.260 \leq P_{\text{patient}} < 0.593 \\ 0, & \text{if } P_{\text{patient}} < 0.260 \end{cases} \tag{30}$$

This framework ensures an optimal balance between sensitivity, specificity, and interpretability by combining probability-based thresholds with structured decision rules for uncertain cases. The next section presents its application to patient-level classification performance.

D. PATIENT-LEVEL CLASSIFICATION PERFORMANCE

To achieve a balanced trade-off between sensitivity and specificity, we adopted a hybrid approach for final patient-level classification. This method integrates probability-based soft voting with an optimized cutoff threshold to enhance robustness. Mathematically, maximizing Youden's Index determined the optimal threshold $T^*_{patient}$, balancing sensitivity and specificity (Equation 15).

As patient cases continued to accumulate during our experimental phase, we initially conducted tests on a limited dataset consisting of 6 IDA cases and 6 THL cases. However, by the time this report was prepared, we had acquired 12 additional THL cases, expanding the total number of test cases to 24 (6 IDA, 18 THL). This expansion aimed to improve model interpretation and robustness.

Patient classification on the expanded test set was conducted using the hybrid approach, which integrates mean-based and argmax-based probability voting, as defined in Equation 30. Among the 24 test cases, only one false positive was observed (Patient No. 3, with an argmax-based probability of 0.553), as detailed in Table 7. Additionally, one false negative was identified in each classification method: the mean-based approach (probability = 0.550) and the argmax-based approach (probability = 0.566), both occurring in Patient No. 23.

The performance of all three classification approaches was comparable, with each method misclassifying only one patient. Despite this overall similarity, we advocate for the hybrid classification framework in this study. This is because it enhances robustness and adaptability in patient-level classification, notably demonstrating superior sensitivity (1.00) and F1-score (0.97) for THL patients compared to IDA patients (sensitivity: 0.83, F1-score: 0.91) (see Table 8).

The model achieved an accuracy of 0.96 on the unseen test set, with detailed classification metrics provided in Table 8 and additional performance visualizations in Figure 5. The model also attained an AUROC of 0.92 for both micro-average and macro-average, with balanced sensitivity and specificity, each measured at 0.92.

Interestingly, results using a fixed 0.5 threshold were identical to those obtained with both the mean-based and argmax-based patient-level classification approaches. This suggests that the hybrid framework naturally converges to a decision boundary of 0.5 for this dataset.

However, the unseen test set consisted of only 24 cases (6 IDA, 18 THL), representing approximately 19% of the total dataset and sourced exclusively from a single hospital. While the hybrid approach performed comparably to a fixed-threshold model on this dataset, further validation on larger, more diverse datasets is necessary to confirm its generalizability. Given its adaptability, mathematical optimization, and clinical interpretability, the hybrid classification approach was selected as the foundation for future dataset expansion.

E. STATISTICAL ANALYSIS FOR CLINICAL DECISION-MAKING IN PATIENT-LEVEL

This section presents the statistical analysis of final patient predictions using calibration plots, likelihood ratios (LR), Brier Score, and R^2 for clinical decision-making.

1) CALIBRATION PLOT

A calibration plot (or reliability diagram) visually assesses how well a model's predicted probabilities align with actual outcomes [43]. It helps determine whether the model is overconfident or underconfident, as illustrated in Figure 6.

Isotonic Regression significantly improved calibration, aligning predicted probabilities more closely with observed outcomes than Platt Scaling, which struggled with mid-range values (0.5–0.7).

In Figure 6 (a), the model underestimates THL likelihood for predicted probabilities between 0.0 and 0.45. The observed probability remains at 0, confirming that cases with low predicted probabilities were IDA, with no false negatives (FN) in this range.

At probabilities near 0.6, the observed probability sharply increases, suggesting the model applies a discrete decision threshold. This implies high classification confidence (THL or IDA) but also indicates uncertainty in the mid-range.

For probabilities above 0.7, the observed probability reaches 1.0, meaning all cases in this range were THL. However, the model lacks smooth probability calibration, leading to an abrupt transition between IDA and THL classification. The inability to capture uncertainty effectively in the mid-range (0.5–0.7) suggests potential overfitting,

TABLE 7. Final patient predictions for the test set using mean-based, argmax-based, and hybrid classification.

No.	Patient ID	True Class	Mean Prob_1	Argmax Prob_1	Mean Pred	Argmax Pred	Hybrid Pred	Hybrid Source
1	IDA-016	0	0.113	0.033	0	0	0	Mean-based
2	IDA-044	0	0.474	0.460	0	0	0	Argmax-based
3	IDA-032	0	0.549	0.553	0	0	1	Argmax-based
4	IDA-023	0	0.193	0.094	0	0	0	Mean-based
5	IDA-031	0	0.347	0.267	0	0	0	Argmax-based
6	IDA-033	0	0.350	0.290	0	0	0	Argmax-based
7	THL-040	1	0.799	0.911	1	1	1	Mean-based
8	THL-053	1	0.843	0.930	1	1	1	Mean-based
9	THL-065	1	0.798	0.905	1	1	1	Mean-based
10	THL-048	1	0.754	0.833	1	1	1	Mean-based
11	THL-067	1	0.870	0.956	1	1	1	Mean-based
12	THL-041	1	0.690	0.756	1	1	1	Mean-based
13	THL-068	1	0.906	0.971	1	1	1	Mean-based
14	THL-069	1	0.934	0.991	1	1	1	Mean-based
15	THL-070	1	0.648	0.706	1	1	1	Mean-based
16	THL-071	1	0.881	0.954	1	1	1	Mean-based
17	THL-072	1	0.897	0.965	1	1	1	Mean-based
18	THL-073	1	0.765	0.829	1	1	1	Mean-based
19	THL-074	1	0.719	0.783	1	1	1	Mean-based
20	THL-075	1	0.908	0.979	1	1	1	Mean-based
21	THL-076	1	0.714	0.766	1	1	1	Mean-based
22	THL-077	1	0.699	0.753	1	1	1	Mean-based
23	THL-078	1	0.550	0.566	0	0	1	Argmax-based
24	THL-079	1	0.922	0.981	1	1	1	Mean-based

Patients 1–12 were from the original test set, while patients 13–24 were additionally included. Given the superior sensitivity (1.00) and F1-score (0.97) for THL prediction, the hybrid classification approach was chosen as the foundation for future dataset expansion.

TABLE 8. Classification report for the unseen test set using the hybrid approach.

Class	Precision	Recall	F1-Score	Support
0 (IDA)	1.00	0.83	0.91	6
1 (THL)	0.95	1.00	0.97	18
Accuracy		0.96 (24 samples)		
Macro Avg	0.97	0.92	0.94	24
Weighted Avg	0.96	0.96	0.96	24

possibly due to factors like data variability (e.g., arising from the single data source with imbalanced classes).

To address this, we applied Platt Scaling (logistic regression-based calibration) and Isotonic Regression to refine predicted probabilities and reduce overconfidence in high probability ranges (see Figure 6 (b)).

Isotonic Regression provided a more flexible, non-parametric mapping that better captured the relationship between predicted and observed probabilities. In contrast, Platt Scaling, which assumes a sigmoid relationship, struggled in the mid-range (below 0.6 and above 0.7), leading to continued overconfidence.

Thus, Isotonic Regression proved to be the more effective calibration method for this dataset, making it a reliable choice for clinical screening, particularly where well-calibrated probability estimates are critical.

2) LIKELIHOOD RATIOS AND BAYESIAN PROBABILITY UPDATES

Likelihood ratios (LR) quantify the impact of a test result on disease probability [44]. The positive likelihood ratio (LR^+) measures how much more likely a positive test result is in a diseased patient compared to a non-diseased one, while the negative likelihood ratio (LR^-) indicates how much more likely a negative test result is in a healthy individual [45].

$$LR_j^+ = \frac{\text{Sensitivity}_j}{1 - \text{Specificity}_j}, \quad j \in \{0, 1\} \tag{31}$$

$$LR_j^- = \frac{1 - \text{Sensitivity}_j}{\text{Specificity}_j}, \quad j \in \{0, 1\} \tag{32}$$

Bayesian probability updates enhance diagnostic certainty by integrating pre-test probabilities with likelihood ratios (LR) [45]. In this study, a pre-test probability of 40% was applied, derived from the estimated THL trait prevalence of 30% to 40% in the Thai population [3].

$$\text{Post-test Odds}_j = \text{Pre-test Odds}_j \times LR_j, \quad j \in \{0, 1\} \tag{33}$$

$$\text{Post-test Probability}_j = \frac{\text{Post-test Odds}_j}{1 + \text{Post-test Odds}_j}, \quad j \in \{0, 1\} \tag{34}$$

Table 9 presents the patient-level diagnostic performance of the hybrid model for identifying IDA and THL, incorporating likelihood ratios and Bayesian post-test probability

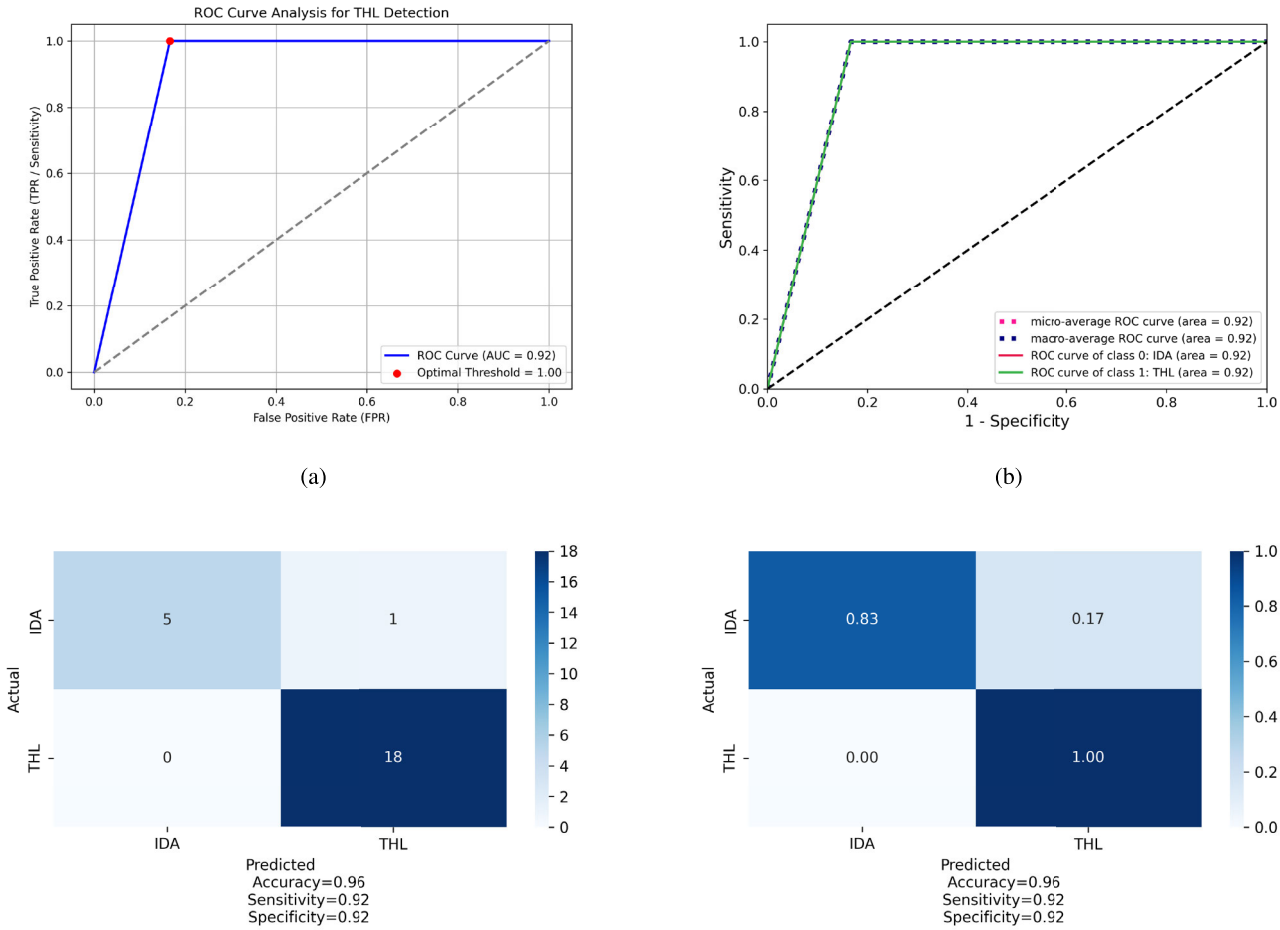


FIGURE 5. **(a) ROC curve analysis for THL detection of the test set, showing an AUROC of 0.92 with an optimal threshold of 1.00. (b) ROC curve analysis for each class detection. (c) Confusion matrix showing classification performance with the number of instances for each class. (d) Confusion matrix showing normalized classification performance for each class.**

updates. The metrics include 95% confidence intervals (CIs) for sensitivity and specificity.

For IDA, a positive test result confirms the condition with near certainty (Post-test Probability ≈ 1.0), while a negative test suggests further iron studies (e.g., serum ferritin) before ruling out IDA.

Supporting this, the model demonstrates high diagnostic certainty. Sensitivity is 83.3% [95% CI : 44%, 97%], indicating good ability to detect IDA when present, and specificity is 100% [95% CI : 82%, 100%], ensuring no false positives. These metrics yield a very high positive likelihood ratio (LR^+) of 8.33×10^9, resulting in a post-test probability of 0.99999999982. The negative likelihood ratio (LR^-) is 0.17, suggesting that a negative result may still require further investigation (e.g., serum ferritin test) before confidently ruling out IDA.

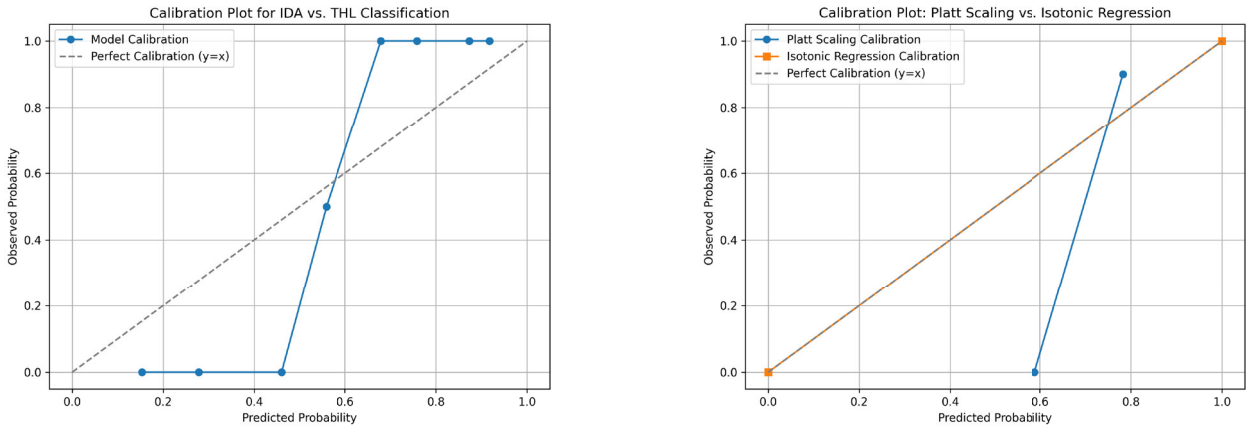
For THL, the model achieves perfect sensitivity of 100% [95% CI : 82%, 100%], indicating all true THL cases are correctly identified. Specificity is 83.3% [95% CI : 61%, 94%], indicating some false positives may occur. The corresponding LR^+ is 6.0, with a post-test probability of 0.80 following a positive result. A negative LR of 0.0 means that a negative result effectively rules out THL in this cohort.

3) BRIER SCORE AND R^2

The Brier Score measures the mean squared error between predicted probabilities and actual labels as defined in Equation 35 [46], where:

$$Brier_{Score} = \frac{1}{N}\sum_{i=1}^{N}(P_{patient,i} - Y_{patient,i})^2 \tag{35}$$

- $P_{patient,i} \rightarrow$ Patient-level final probability (derived from Equation 30)
- $Y_{patient,i} \rightarrow$ True label for patient i (0 for IDA, 1 for THL)
- $N \rightarrow$ Total number of patients.



(a) (b)

FIGURE 6. Comparison of calibration plots. (a) For the default plot, the model's calibration line implies high classification confidence (for THL or IDA predictions) but indicates uncertainty in the mid-range. (b) For Platt Scaling (LR-based calibration) versus Isotonic Regression plots, the Isotonic Regression line indicates a more effective calibration method for this dataset when applied to the hybrid approach's output.

TABLE 9. Patient-level performance evaluation and Bayesian probability update using LR for IDA and THL including 95% confidence intervals for sensitivity and specificity.

Metric	IDA	THL
TP	5	18
TN	18	5
FP	0	1
FN	1	0
Sensitivity	0.8333	1.0
95% CI of Sensitivity	[0.44, 0.97]	[0.82, 1.0]
Specificity	1.0	0.8333
95% CI of Specificity	[0.82, 1.0]	[0.61, 0.94]
LR^+	8.33×10^9	6.0
LR^-	0.1667	0.0
Post-test Probability	0.99999999982	0.8

Note: Post-test probability uses Bayes' Theorem with prior $P = 0.4$

A lower Brier Score, closer to 0, indicates better probability calibration, while a score near 1 suggests poor reliability.

The R^2 Score evaluates how well predicted probabilities explain the variance in true outcomes by Equation 36 [47], where $\bar{Y}_P$ is mean of true labels.

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(P_{patient,i} - Y_{patient,i})^2}{\sum_{i=1}^{N}(Y_{patient,i} - \bar{Y}_{patient})^2} \tag{36}$$

Table 10 presents the patient-level evaluation for clinical interpretation using an ensemble model of EfficientNetV2S. Evaluation A utilized the original testing set (12 cases), while Evaluation B expanded to include 6 cases of IDA and 18 cases of THL, totaling 24 cases (from Table 7).

The Brier Scores of 0.0834 (A) and 0.0697 (B) indicate that the predicted probabilities are well-calibrated, meaning

TABLE 10. Patient-level model evaluation and clinical interpretation.

Metric	Evaluation A	Evaluation B
Patient Distribution	6 (IDA), 6 (THL)	6 (IDA), 18 (THL)
Brier Score*	0.0834	0.0697
R^2 Score**	0.6663	0.6284
Screening Reliability	High	High
Decision Support	Yes	Yes
Cost-effectiveness	Yes	Yes

The optimal thresholds were utilized. *Lower is better, **Higher is better.

the model's confidence in its predictions aligns closely with actual outcomes. Lower Brier Scores suggest that the model effectively represents uncertainty in classification, making it suitable for risk estimation and clinical decision support.

The R^2 Scores of 0.6284 and 0.6663 suggest that the model explains 62.84 to 66.63% of the variance in predicted patient classifications, which indicates strong discriminatory power between IDA and THL cases. While R^2 is more commonly used in regression, its use here suggests that the model's probability predictions correlate well with actual outcomes.

Given these results, the model is well-suited for initial screening before confirmatory laboratory tests—such as serum ferritin for IDA and hemoglobin electrophoresis for THL. Its well-calibrated probabilities and discriminatory strength support its role in assisting clinicians with early detection.

The combination of low Brier Scores and high R^2 Scores provides strong evidence of their effectiveness in distinguishing THL from IDA at the patient level. Consequently, this approach supports early diagnosis with high classification accuracy, positioning the models as an AI-assisted screening tool that aids hematologists in prioritizing at-risk patients for

further testing, thereby improving efficiency in diagnostic workflows.

F. MODEL BEHAVIOR ANALYSIS

Model behavior was analyzed using SHAP and Grad-CAM on EfficientNetV2S, trained on the fold-4 dataset (overall accuracy: 0.80). Figure 7 illustrates the model's interpretability in detecting four unseen cases of IDA and THL from the test set.

Our analysis revealed that misclassifications—both false positives and false negatives—were primarily driven by edge-based features, rather than core morphological characteristics, indicating limitations in feature representation.

In cases (1) and (2), with soft-voting accuracies of 0.75 and 0.74, respectively, SHAP visualizations showed no clear discrimination between blue-negative and red-positive areas, reflecting model uncertainty. False-positive patches in case (1) appeared in regions A, D, and F, while false-negative patches in case (2) were observed in regions A and F, suggesting reliance on non-discriminative features, likely due to data imbalance causing feature attribution overlap.

Grad-CAM analysis further demonstrated that misclassified regions were often situated near cell edges, where predictions were influenced by incomplete cell shapes, adjacent boundaries, or staining artifacts rather than meaningful cellular texture. Additionally, noise introduced during staining normalization contributed to contextual degradation, further limiting patch-based classification accuracy. These findings highlight the need for more robust, edge-preserving preprocessing techniques within the patch extraction pipeline. Notably, even for expert hematologists, such cases remain diagnostically challenging.

To further evaluate model interpretability, we examined correctly classified cases (3) and (4), with soft-voting accuracies of 0.96 and 0.93 for IDA and THL, respectively. SHAP visualizations showed distinct blue-negative areas for IDA in (3) and stronger red-positive tones for THL in (4). Grad-CAM results confirmed that the most activated regions were concentrated at the center of cells, particularly for THL cases.

In comparison, SHAP provides pixel-level attribution maps, whereas Grad-CAM offers a broader view of spatial attention. Despite the strong overall performance, the model's behavior suggests a degree of overfitting to specific low-level features.

Rather than explicitly segmenting regions of interest (ROI), we implemented an optimized preprocessing pipeline designed to enhance feature stability and support practical deployment for our case study hospital. This pipeline includes blood smear normalization in the LAB color space, patch extraction at the cell level (via Algorithm 1), and stratified multi-fold cross-validation with data augmentation.

G. COMPARISON TO RELATED WORKS

Previous studies [18], [19] [20] have explored DL for hematologic disorders, particularly in medical imaging-based classification. However, our study differentiates itself through a structured hybrid approach that integrates soft voting with optimized probability thresholds to enhance patient-level classification.

While Zaylaa et al. [18] and Lian et al. [19] focused on MRI-based segmentation for hematologic assessment, our method applies hybrid AI classification directly to blood smear images, making it more pertinent for clinical hematology screening. In blood smear-based classification, our approach outperformed Wiratchawa et al. [20], achieving higher accuracy (96% vs. 86%) and F1-score (0.94 vs. 0.85).

Our study bridges the gap between blood smear-based AI classification and clinically relevant patient-level decision-making by integrating probability thresholding, Bayesian updates, and model explainability.

V. LIMITATIONS AND FUTURE WORK

Despite demonstrating strong performance, several limitations must be addressed prior to real-world clinical deployment.

First, although our original test set comprised only 12 cases (6 IDA, 6 THL), we subsequently expanded it by incorporating an additional 12 THL cases, resulting in a total of 24 cases (6 IDA, 18 THL). This expansion aimed to enhance interpretability and facilitate a more robust evaluation. Nevertheless, the limited size and class imbalance of the test set raise concerns regarding overfitting and generalizability. While our hybrid method exhibited promising results, its performance was similar to that of a fixed 0.5 threshold on this dataset. This suggests its reliability may not consistently extend to larger, more diverse cohorts. Future studies should validate the model across multi-center datasets to ensure consistent behavior and clinical reliability.

Second, class imbalance within the training data (IDA: 38; THL: 61) may influence model learning. Despite our mitigation strategies, the model may favor features associated with THL, potentially introducing bias upon deployment in broader clinical settings. A more balanced and representative dataset is essential to improve fairness and generalization. Techniques such as Synthetic Minority Over-sampling Technique (SMOTE) [48] and weighted loss functions [49], widely used for addressing class imbalance, will be explored in future work to mitigate this issue.

Third, model calibration remains an area requiring improvement. The likelihood ratios ($LR^{+} = 8.33 \times 10^{9}$, $LR^{-} = 0.1667$) were mathematically extreme, likely reflecting the limited sample size and RBC morphology distribution. Although Platt scaling and isotonic regression were applied, clinical uncertainties such as biological variation and image quality may still compromise reliability. Larger, more diverse datasets are necessary to validate and refine these estimates.

Lastly, the recall performance for IDA was lower than that for THL, which may limit its clinical utility. Increasing the number of IDA cases is a key priority. While five-fold cross-validation with a 0.25 holdout ratio supports

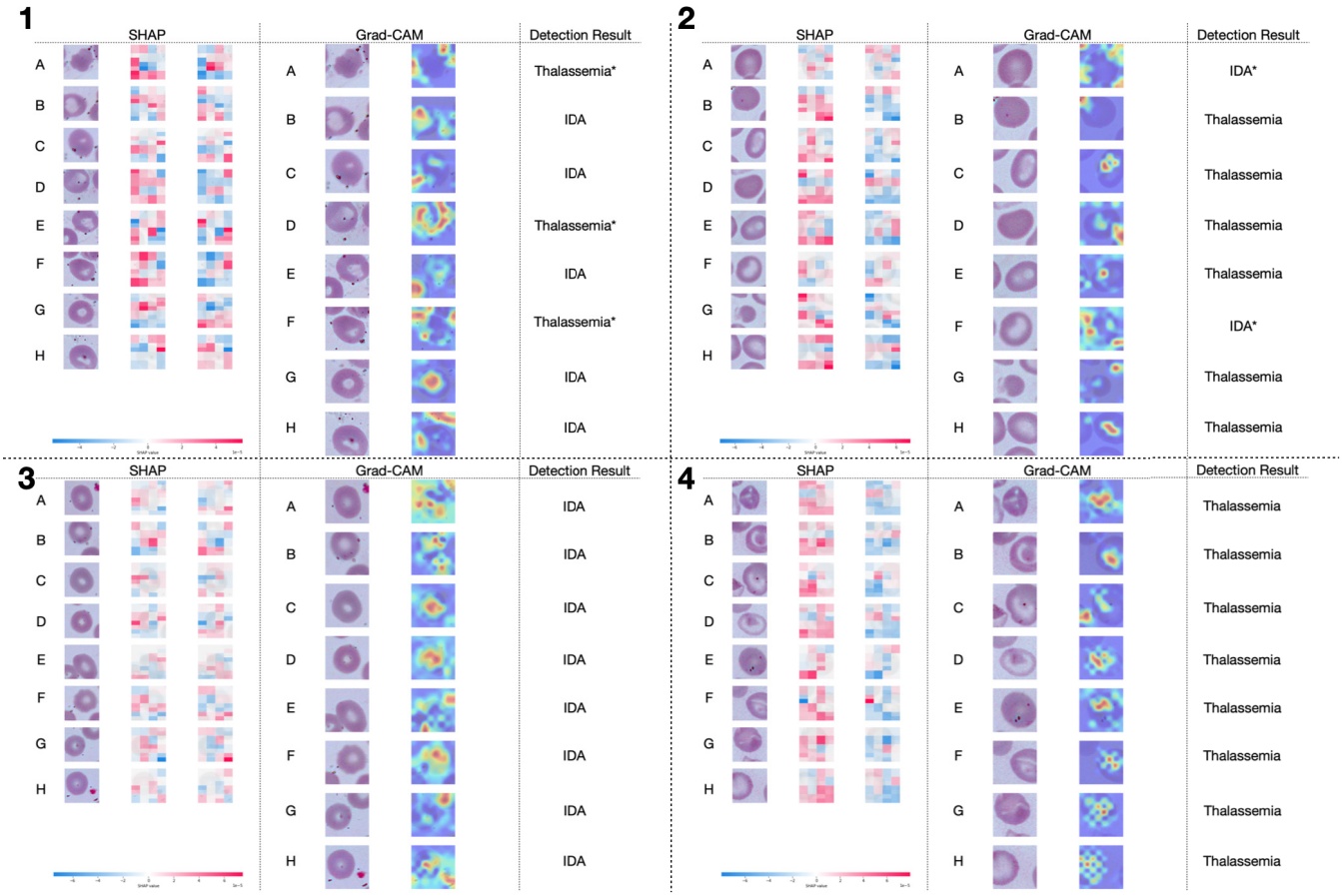


FIGURE 7. Model behavior analysis of EfficientNetV2S trained on the fold-4 dataset using SHAP and Grad-CAM for detecting four unseen cases of IDA and THL from the testing set. The ground truth labels are as follows: (1) and (3) correspond to IDA, while (2) and (4) correspond to THL. The soft-voting accuracies are (1) 0.75 (where * denotes FP), (2) 0.74 (where * denotes FN), (3) 0.96, and (4) 0.93.

TABLE 11. Comparison of AI-based THL screening approaches.

Study	Method	Data Type	Performance	Patient-Level Decision
Zaylaa et al. [18]	AI-based MRI segmentation	MRI scans	Accuracy = 88%	No
Lian et al. [19]	Deep Learning MRI segmentation	MRI scans	Dice: 0.90 (internal), 0.82-0.86 (external)	No
Wiratchawa et al. [20]	CNN-based ThalNet	Blood smear images	Accuracy = 86%, F1-score = 0.85	No
Our Work	Hybrid AI with probability thresholds	Blood smear images	Accuracy = 96%, F1-score = 0.94	**Yes**

generalization, the full dataset (99 training cases) remains below the preferred minimum of 150 cases per class. Further validation with external, multi-source datasets is essential for establishing model consistency and scalability.

Despite these limitations, this study establishes a practical and interpretable AI-based framework for anemia screening, providing a strong foundation for future development and deployment.

VI. CONCLUSION

IDA and THL are prevalent hematologic disorders necessitating precise and efficient screening for early diagnosis and informed clinical decision-making. Traditional manual blood smear analysis is time-consuming, subjective, and lacks reproducibility, highlighting the urgent need for AI-driven solutions that effectively balance sensitivity, specificity, and interpretability.

This study introduces a novel hybrid patient-level classification framework that integrates soft voting with an optimized probability-based thresholding strategy. Unlike traditional approaches that often rely on single-model outputs or lack explicit calibration for clinical decision support, our framework combines multiple AI strengths to provide robust and interpretable patient-level diagnoses. Rather than focusing solely on performance comparison, our approach establishes a structured decision-making framework that enhances model robustness and applicability. Validated using a real-world dataset from Hatyai Hospital, Thailand, the model was assessed at both patch and patient levels to ensure reliability for deployment in clinical settings.

Experimental results demonstrated strong classification performance across multiple evaluation levels. The model achieved an overall accuracy of 96% on the unseen test set, with precision-recall values of 1.00 and 0.83 for IDA, and 0.95 and 1.00 for THL, respectively. At the patient level, sensitivity for THL detection reached 1.00, while IDA sensitivity was 0.83, ensuring robust identification of positive cases.

Beyond accuracy, statistical validation using Bayesian probability updates reinforced the reliability of predictions, yielding post-test probabilities of 99.999999982% for IDA and 80% for THL. The likelihood ratio negative (LR^-) values of 0.1667 for IDA and 0.0 for THL underscored the model's ability to minimize false negatives, a critical factor in early disease detection. Furthermore, the model explained 62.84% of the variance in predicted patient classifications, demonstrating strong discriminatory power.

A key strength of this study is the explainability of the hybrid classification model. SHAP and Grad-CAM analyses revealed that the deep learning model primarily relied on key morphological features of red blood cells, ensuring transparency and interpretability.

While the proposed hybrid approach provides a structured baseline for AI-driven hematology screening, several challenges remain. The dataset was sourced from a single center (Hatyai Hospital), which may limit generalizability. Additionally, the hybrid cutoff threshold could benefit from adaptive optimization techniques for further refinement. Prospective validation studies are necessary to integrate this system into routine hematology workflows and assess its performance across diverse patient populations.

Overall, this study presents a foundational framework for AI-assisted anemia screening, integrating hybrid probability thresholds, soft voting, Bayesian updates, and explainability techniques. While the proposed model exhibits strong performance in distinguishing IDA and THL, we acknowledge the critical need for additional large-scale, multi-center external validation to confirm its robustness and generalizability. Nonetheless, this structured approach lays the groundwork for significant future advancements in clinical AI-assisted diagnostics, facilitating data-driven refinement, adaptive optimization, and eventual real-world integration across diverse healthcare settings.

ABBREVIATIONS

See Table 12.

DATASET AND CODE AVAILABILITY

The dataset is publicly available at https://dx.doi.org/10.21227/a6df-8755. The code demonstrating the development of the hybrid AI-based patient-level classification framework—integrating soft voting with optimized probability-based thresholds—is available at https://github.com/kasikrit/IDA-THL-Classification/

TABLE 12. List of abbreviations.

Abbreviation	Definition
AI	Artificial Intelligence
DL	Deep Learning
ML	Machine Learning
TL	Transfer Learning
RBC	Red Blood Cell
IDA	Iron Deficiency Anemia
THL	Thalassemia
CBC	Complete Blood Count
TIBC	Total Iron-Binding Capacity
MCV	Mean Corpuscular Volume
MCH	Mean Corpuscular Hemoglobin
MCHC	Mean Corpuscular Hemoglobin Concentration
TPR	True Positive Rate
TNR	True Negative Rate
PPV	Positive Predictive Value
NPV	Negative Predictive Value
LR^+	Positive Likelihood Ratio
LR^-	Negative Likelihood Ratio
SHAP	SHapley Additive exPlanations
Grad-CAM	Gradient-weighted Class Activation Mapping
ROI	Region of Interest

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